Research Statement

Internal Dynamics, Geometry, and Mechanistic Principles of Learning Systems

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My research is driven by a simple question: can a neural network be understood not only through the function it computes, but through the internal principles that organize its computation? I am interested in what happens between input and output—how hidden states evolve, how representations move through learned spaces, how different inputs alter the pace of internal computation, and why some internal trajectories are associated with reliable predictions while others accompany failure.

Physics strongly influences how I formulate these questions. I do not assume that neural networks literally obey physical laws, nor do I use physical analogies as decoration. Instead, physics provides a disciplined vocabulary for asking structural questions about learning systems: What constitutes a state? What does it mean for that state to change? Can a model possess an intrinsic notion of time? Does geometry affect how internal evolution should be measured? Can learned dynamics be organized around an optimization or variational principle? And can such a principle be falsified through controlled interventions?

This perspective has produced a sequence of increasingly mechanistic questions throughout my master's research. In TST-Mamba, I studied state-space recurrence as selective accumulation and compression of temporal change. In RiemannMamba, I introduced geometry-aware internal time to measure how strongly an input drives hidden-state evolution and investigated its relationship with model difficulty and failure. My current work asks a deeper question: whether the learned state trajectories of selective state-space models themselves exhibit a systematic variational structure under an input-dependent internal clock.

These projects share the same underlying goal. I want to turn internal quantities of neural networks into experimentally testable objects and use them to formulate hypotheses about how learned computation is organized. Ultimately, I am interested in a mechanistic understanding of learning systems: not only identifying internal patterns that correlate with behavior, but determining which structures are functional, how they emerge through training, and what changes internally when a model fails.

1 Research Conducted During My Master's Program

1.1 TST-Mamba (Under review, Neurocomputing)

State Evolution as Selective Compression of Spatiotemporal Change

Abstract

Non-Line-of-Sight (NLOS) imaging reconstructs hidden scenes from time-resolved scattered optical measurements, but photon-limited SPAD transients are often degraded by low signal-to-noise ratios, temporal broadening, and background noise, which can substantially reduce downstream reconstruction quality. In this paper, we propose TST-Mamba, a plug-and-play denoising model for NLOS transient data based on selective state-space modeling. The proposed method follows a Temporal–Spatial–Temporal design: a temporal Mamba encoder models patch-wise transient dynamics, a spatial Mamba block aggregates context from compact bottleneck tokens, and a skip–latent fusion decoder restores the denoised transient volume while preserving temporal structure. This design can be interpreted as a Mamba-based skip-connected denoising autoencoder, where spatial context is processed in the compressed latent space instead of repeatedly mixing the full spatio-temporal volume. On simulated NLOS transients, TST-Mamba improves PSNR from 23.35 dB to 32.94 dB on the Seen split and increases SSIM from 0.0818 to 0.8807 on the Unseen split. It also consistently enhances downstream reconstruction across multiple NLOS pipelines and generalizes to real measurements, demonstrating its effectiveness as a plug-and-play transient-domain denoising module for practical NLOS imaging.

I began TST-Mamba from a question about temporal representation. A sequence is indexed by external time, but its information lies in change: what differs between consecutive observations and which of those changes remain relevant to future computation. This viewpoint was especially natural for non-line-of-sight (NLOS) imaging, where a hidden scene is reconstructed from time-resolved photon measurements after indirect light transport.

A selective state-space model evolves through a recurrence of the form

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \qquad y_t = C_t h_t.$$

Rather than viewing the hidden state simply as the representation located at the final sequence position, I interpreted it as an accumulated record of changes that the model selectively retains. Information that repeatedly contributes to subsequent state evolution persists, whereas less useful variation can be compressed away.

This interpretation guided the architecture of TST-Mamba. Temporal Mamba first processes the photon sequence independently at each spatial location and compresses its temporal evolution into a latent state. Spatial Mamba then models relationships among these latent spatial representations. Because aggressive temporal compression can remove weak but reconstruction-critical photon responses, I introduced a skip pathway that preserves fine temporal evidence and reinjects it after latent spatial processing.

I also developed a real-measurement-guided SPAD noise simulator and evaluated the resulting denoiser across simulated, unseen-category, and real NLOS measurements. The project taught me an important lesson that has continued to shape my research: architectural choices become more interpretable when they are derived from an explicit hypothesis about what the internal state is doing, rather than treated only as components to be optimized empirically.

1.2 RiemannMamba (Under review, NeurIPS 2026)

Geometry-Aware Internal Time and White-Box Reliability

Abstract

Output confidence is widely used for selective prediction and reliability estimation, but it is a post-hoc signal computed only from the final prediction distribution and does not directly reveal how a model internally evolves while processing an input. In Selective State Space Models, this limitation is especially important because the input-dependent time step Δ determines the discretized state transition that governs hidden-state evolution, suggesting that reliability-relevant information may already be present in the model's internal dynamics. In this work, we propose RiemannMamba, which exposes such dynamics as an internal time variable τ. Rather than treating Δ only as a computational gate, we use it to construct per-dimension metric weights on the hidden-state space, so that each hidden-state update is measured as a geometry-weighted change and accumulated into an internal time variable τ. We further introduce a difficulty-alignment objective that encourages τ to reflect sample difficulty while keeping it grounded in the hidden-state trajectory. Experiments on ImageNet-100 and CIFAR-100 show that τ becomes strongly aligned with input difficulty and error propensity. Under distribution shift, τ-based scoring achieves lower AURC than confidence-only scoring across 75 corruption scenarios, reducing AURC by 2.7% on ImageNet-100-C and 7.6% on CIFAR-100-C relative to confidence-only scoring. These results suggest that Selective SSMs contain an internal reliability signal derived from hidden-state evolution, and that geometry-aware internal time provides a principled perspective for reliability estimation from model dynamics.

TST-Mamba led me to ask whether equal external sequence steps necessarily correspond to equal amounts of internal computation. Two inputs may pass through exactly the same architectural depth while inducing very different hidden-state changes. One may be processed with little reorganization of the state, while another may produce a much larger internal transition.

This motivated RiemannMamba. Drawing on the structural idea of intrinsic time in geometry, I asked whether model evolution could be measured using a notion of duration determined jointly by the trajectory of the hidden state and an input-dependent local geometry. Mamba provides a natural quantity for this purpose through its input-dependent discretization step Δ. Rather than treating Δ only as a selective gate, I used it as a geometric factor for constructing an input-dependent metric over hidden-state changes.

For hidden-state transitions

$$\Delta h_k = h_{k+1} - h_k,$$

I defined geometry-aware internal time as

$$\tau = \sum_k \sqrt{\Delta h_k^\top g_k \Delta h_k},$$

where g_k is an input-dependent metric derived from the selective dynamics of the model. The resulting quantity measures the accumulated amount of geometry-weighted hidden-state evolution induced by an input.

Experimentally, I found that inputs processed through the same architecture could produce substantially different internal times. Larger τ values were associated with difficult, error-prone, corrupted, and distribution-shifted inputs. This made internal time useful not only as a description of state evolution, but as a complementary white-box reliability signal for selective prediction.

I further developed τ-CAM, a gradient-free method that decomposes accumulated internal time spatially and identifies image regions associated with stronger hidden-state evolution. This project shifted my focus from designing architectures toward studying the internal observables of learned systems. More importantly, it raised the next question: if internal time systematically reflects how a model processes an input, is that time merely an observable, or is the learned trajectory itself organized according to a deeper dynamical principle?

1.3 Ongoing Research: Variational Structure of Internal Time in Mamba

From an Internal Clock to Stationary Trajectories and Mechanistic Hypotheses

My current research investigates whether the learned state trajectories of selective state-space models possess a systematic structure when measured under their input-dependent internal clock. The central idea is to treat Mamba not simply as a sequence of feature transformations, but as a dynamical system that transports representations through state space.

For each position t and channel d, Mamba generates a positive input-dependent discretization interval

$$\Delta(t,d) = \text{softplus}\left(\widetilde{\Delta}(t,d) + b_{\Delta,d}\right).$$

Accumulating this quantity produces an internal progression coordinate. My first question is whether this internal time contains information genuinely distinct from geometric state-space distance. I therefore compare accumulated internal time with the length of the corresponding hidden-state trajectory, including matched-distance comparisons and cases in which a geometrically longer trajectory nevertheless uses less internal time.

The more fundamental question is whether a learned trajectory is special with respect to a travel-time functional. Because Δ is generated from the input and model computation rather than given directly as a function of state, I first estimate a local internal velocity field from observed state transitions and internal-time increments. This allows a candidate trajectory γ to be assigned an internal travel time of the form

$$T[\gamma] = \int_\gamma \frac{ds}{v_{\text{int}}(S, \text{context})}.$$

I then ask whether the trajectory actually produced by a trained Mamba model is close to a stationary point of this functional,

$$\delta T[\gamma_{\text{Mamba}}] \approx 0,$$

and, more strongly, whether it behaves as a local minimum under endpoint-preserving perturbations,

$$T[\gamma_{\text{Mamba}}] \leq T[\gamma_{\text{Mamba}} + \epsilon\eta], \qquad \eta(0) = \eta(1) = 0.$$

This formulation is deliberately falsifiable. A stationary trajectory need not be a minimum, and an apparent minimum-time pattern need not arise from Mamba's selectivity. I therefore separate several hypotheses rather than assuming a single principle in advance.

First, I test whether internal time is genuinely non-redundant with geometric path length. Second, I test stationarity and local minimality using controlled trajectory perturbations and alternative paths. Third, I examine the causal role of selectivity by comparing the learned input-dependent Δ with fixed, shuffled, and randomized alternatives. Fourth, I track the same quantities across training checkpoints to determine whether any variational structure is present at initialization or emerges through learning.

I also intend to test whether the phenomenon survives architectural changes, including the different discretization scheme of Mamba-3. If it does, this would suggest a principle broader than a particular implementation detail; if it does not, the failure would itself identify the architectural conditions under which the interpretation is valid.

Finally, I am interested in the relationship between violations of the proposed dynamical structure and model failure. If misclassified, corrupted, or out-of-distribution inputs systematically produce larger stationarity residuals or optimality gaps even after controlling for output confidence, these quantities could provide a white-box diagnostic of atypical internal computation.

The broader aim of this project is not to claim that neural networks obey a pre-existing physical law. It is to ask whether a trained model exhibits an empirical organizing principle that can be stated mathematically, challenged with counterexamples, and tested through intervention. This is the direction in which I want my research to continue.

2 Future Research

From Data and Internal Observables to Mechanistic Principles of Learned Computation

My future research will focus on understanding neural networks through the structures that organize their internal computation and the data that shape those structures during learning. My training in data science has made me particularly interested in treating data not merely as inputs used to optimize a model, but as one of the forces that determines what internal representations and dynamics eventually emerge. I am especially interested in four connected problems: discovering interpretable internal principles, distinguishing causal mechanisms from correlations, understanding how training data shape those mechanisms, and using internal dynamics to study reliability and failure.

2.1 Discovering Organizing Principles in Learned Dynamics

Modern neural networks contain enormous numbers of internal variables, but description alone does not produce understanding. Recording activations, visualizing representations, or identifying correlated directions can reveal patterns without explaining why those patterns arise.

I want to search for lower-dimensional principles that organize these high-dimensional dynamics. Internal time is one example: instead of treating every hidden-state transition independently, it asks whether many transitions can be understood through a common notion of progression. A variational principle would go further by asking whether the trajectories themselves are organized relative to a measurable objective.

I am interested in extending this style of analysis beyond a single quantity or architecture. Possible objects include characteristic time scales, trajectory curvature, contraction and expansion of representations, dynamical bottlenecks, conserved or approximately invariant quantities, and transitions between functional state-space regions.

The important criterion is falsifiability. A useful mechanistic principle should make predictions about what happens when the model, input, or internal state is perturbed. It should also specify conditions under which the proposed interpretation fails.

2.2 From Correlation to Causal Mechanism

A recurring challenge in interpretability is that an internal quantity can be strongly correlated with model behavior without playing a functional role in producing that behavior. I therefore want causal testing to be central to my research.

My current work already moves in this direction by modifying Mamba's input-dependent selectivity and observing whether the proposed trajectory structure survives. More generally, I want to use interventions on internal states, dynamical parameters, representation directions, and computational time scales to distinguish quantities that merely accompany model behavior from mechanisms that actively contribute to it.

This also requires appropriate controls. Internal signals should be compared against matched-confidence examples, randomized or counterfactual interventions, alternative architectures, and models observed throughout training. A proposed mechanism becomes substantially more convincing when modifying it produces predictable changes in behavior while plausible confounders remain controlled.

2.3 From Discrete Training Data to Continuous Learned Structure

My interest in the role of data was initially inspired by the contrast between discrete observations and the continuous spaces in which they are embedded. A training dataset contains only a finite collection of observed examples, while the space of possible inputs and internal representations is vastly larger and effectively continuous. This led me to ask a basic question: what structure does learning create in the regions that are not directly occupied by training samples?

I often think about this question through a field-based analogy from physics. Individual masses do not influence only the points they occupy; collectively, they generate a gravitational field throughout the surrounding space. I do not assume that training samples literally behave as physical masses, but the analogy provides a useful structural question: can discrete training examples collectively shape a learned field that extends beyond the observations themselves and organizes how previously unseen inputs are represented and processed?

From this perspective, the effect of data should not be visible only in the parameters or final predictions of a trained model. Different examples, densities, class configurations, and atypical samples may alter the geometry of representation space, the directions of internal state evolution, and the local dynamics encountered by new inputs. A new sample would then not be processed in isolation; its internal trajectory would depend partly on the structure that the training data have already induced throughout the learned space.

I want to make this idea experimentally testable. One approach is to remove, reweight, subsample, or perturb selected training examples and examine how the resulting representation geometry and internal trajectories change. If a specific subset of data is responsible for a particular internal structure, modifying that subset should produce a predictable deformation of the learned geometry or dynamics. Data-attribution and influence-based methods could help identify candidate examples, while retraining and controlled intervention could test whether their effects are genuinely causal.

This perspective connects my background in data science with my interest in mechanistic interpretability. Rather than asking only which training examples influence a final prediction, I want to understand how discrete observations collectively construct the internal space through which future inputs are processed. Ultimately, I am interested in tracing a path from training data, to learned geometry and dynamics, and finally to generalization, reliability, and model failure.

2.4 Understanding Reliability from the Inside Out

RiemannMamba showed me that output confidence does not exhaust the information available inside a model. Inputs with similar confidence can induce different amounts of internal evolution, and difficult or shifted samples can reveal themselves through their internal dynamics before the final prediction is examined.

I want to develop this idea further by studying whether trajectory-level quantities provide complementary information about model reliability. Potential signals include atypical internal time, deviations from learned trajectory structure, abrupt transitions between representation regions, sensitivity of internal states to small perturbations, and disagreement among layers or computational time scales.

I am also interested in whether reliability failures can be traced backward to the data conditions under which the relevant internal mechanisms were formed. If particular failure modes repeatedly arise from identifiable training-data regimes or influential subsets, reliability analysis could move beyond detecting unusual behavior toward explaining how that behavior was learned.

The goal is not merely to construct another uncertainty score. I want to understand why a particular internal pattern is associated with failure and whether manipulating that pattern, or the data that produced it, changes the failure itself. Reliability analysis and mechanistic interpretability should therefore reinforce one another: reliability provides behavior that needs explanation, while internal mechanisms provide hypotheses about where that behavior originates.

2.5 Beyond Mamba: Toward Generalizable and Reliable AI

Selective state-space models are especially useful for my current research because their recurrence, discretization, and hidden-state dynamics make internal evolution explicit. However, I do not view Mamba as the final scope of my research, but as a tractable system in which questions about time, state evolution, trajectory structure, and model failure can be tested under controlled conditions.

A central long-term question for me is what allows learned mechanisms to remain stable beyond the conditions seen during training. A model may perform well on familiar data while relying on internal structures that become fragile under distribution shift or novel inputs. In contrast, a genuinely generalizable model should preserve functionally meaningful internal organization across diverse inputs and environments.

I therefore want to identify which internal principles remain stable across architectures, datasets, and distributional changes, and which break when generalization fails. My goal is to move from tractable systems toward larger and more complex models while preserving controlled experimentation, and to use mechanistic understanding to help develop AI systems that behave reliably across diverse and previously unseen conditions.